

# Презентация лабораторной работы

## Лабораторная работа №3

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# Раздел 1

## Информация

- **Швецов Михаил Романович**



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## Раздел 2

### Выполнение лабораторной работы

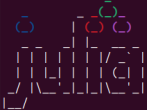
Создаём каталог лабораторной работы. (рис. 1).

```
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab0
3
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03$ j
ulia

    Info The latest version of Julia in the `release` channel is 1.12.7+0.x64.linux.gnu
. You currently have `1.12.5+0.x64.linux.gnu` installed. Run:

    juliaup update

in your terminal shell to install Julia 1.12.7+0.x64.linux.gnu and update the `release` cha
nnel to that version.
```



```
Documentation: https://docs.julialang.org

Type "?" for help, "]"? for Pkg help.

Version 1.12.5 (2026-02-09)
Official https://julialang.org release

julia> using Pkg

julia> Pkg.add("DrWatson")

Warning: could not download https://pkg.julialang.org/registries
exception = RequestError: Failed to connect to 127.0.0.1 port 10809 after 0 ms: Could n
ot connect to server while requesting https://pkg.julialang.org/registries
 @ Pkg.Registry ~/.julia/juliaup/julia-1.12.5+0.x64.linux.gnu/share/julia/stdlib/v1.12/Pkg
/src/Registry/Registry.jl:83
Updating registry at ~/.julia/registries/General.toml
Resolving package versions...
Project No packages added to or removed from ~/.julia/environments/v1.12/Project.toml
Manifest No packages added to or removed from ~/.julia/environments/v1.12/Manifest.toml

julia> using DrWatson

julia> initialize_project("project"; authors="Швецов Михаил", git=false)
Activating new project at ~/2026-1--study--simulation-modeling/labs/lab03/project
```

Далее в каталоге src создаём необходимый скрипт (рис. 2).

```
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project$ cd src
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/src$ nano daisyworld.jl
```

Рисунок 2: Создание скрипта в необходимом каталоге:

Устанавливаем все необходимые пакеты (рис. 3) (рис. 4).

```
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$ cd ../
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. -e 'using Pkg; Pkg.add(["Agents", "DataFrames", "Plots"])'
Updating registry at `~/.julia/registries/General.toml`
Resolving package versions...
Installed Statistics — v1.11.5
Installed SimpleTraits — v0.9.6
Installed OnlineStatsBase — v1.7.3
Installed StructArrays — v0.7.3
Installed StreamSampling — v0.8.2
Installed Agents — v7.0.3
```

Рисунок 3: Установка:

```
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project$ julia --project=. -e 'using Pkg; Pkg.add("StatsBase")'
Resolving package versions...
Updating `~/2026-1--study--simulation-modeling/labs/lab03/project/Project.toml`
[2913bbd2] + StatsBase v0.34.13
Manifest No packages added to or removed from `~/2026-1--study--simulation-modeling/labs/lab03/project/Manifest.toml`
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project$
```

Рисунок 4: Установка:

Создаём и выполняем scripts/daisyworld.jl (рис. 5).

```
...dependencies successfully precompiled in 100 seconds (921 already precompiled)
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
object$ julia --project=. scripts/daisyworld.jl
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
object$
```

Рисунок 5: Выполнение:

# Создаём литературный стиль (чистый код, markdown, notebook) (рис. 6) (рис. 7) (рис. 8).

```
julia> using Literate

julia> Literate.markdown("scripts/daisyworld.jl", "docs")
[ Info: generating markdown page from "~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld.jl"
[ Info: writing result to "~/2026-1--study--simulation-modeling/labs/lab03/project/docs/daisyworld.md"
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/docs/daisyworld.md"
```

Рисунок 6: Markdown:

```
julia> Literate.notebook("scripts/daisyworld.jl", "docs")
[ Info: generating notebook from "~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld.jl"
[ Info: executing notebook 'daisyworld.ipynb'

[ Info: writing result to "~/2026-1--study--simulation-modeling/labs/lab03/project/docs/daisyworld.ipynb"
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/docs/daisyworld.ipynb"
```

Рисунок 7: Notebook:

```
julia> Literate.script("scripts/daisyworld.jl", "docs")
[ Info: generating plain script file from "~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld.jl"
[ Info: writing result to "~/2026-1--study--simulation-modeling/labs/lab03/project/docs/daisyworld.jl"
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/docs/daisyworld.jl"

julia> 
```

Рисунок 8: Script:

Создадим скрипт `scripts/daisyworld-animate.jl` и загрузим его на фидеохостинг RUTUBE (рис. 9).

```
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr  
object$ cd scripts  
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr  
object/scripts$ nano daisyworld-animate.jl
```

Рисунок 9: Создание скрипта

Создадим скрипт scripts/daisyworld-count.jl (рис. 10).

```
object/scripts$ nano daisyworld-count.jl
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
object/scripts$ julia --project=. daisyworld-count.jl
  Activating project at `~/2026-1--study--simulation-modeling/labs/lab03/project`
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
object/scripts$
```

Рисунок 10: Создание и выполнение:

## Создание литературного стиля (рис. 11).

```
julia> Literate.markdown("daisyworld-count.jl", "docs")
^[[D^[[C^[[C[ Info: generating markdown page from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count.md`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count.md"

julia> Literate.script("daisyworld-count.jl", "docs")
[ Info: generating plain script file from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count.jl`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count.jl"

julia> Literate.notebook("daisyworld-count.jl", "docs")
[ Info: generating notebook from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count.jl`
[ Info: executing notebook `daisyworld-count.ipynb`
```

Рисунок 11: Литературный стиль:

Создание файла `scripts/daisyworld-luminosity.jl`, создание для него литературного стиля (рис. 12).

```
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$ nano daisyworld-luminosity.jl
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$ julia
      Info: The latest version of Julia in the 'release' channel is 1.12.7+0.x64.linux.gnu.
      You currently have '1.12.5+0.x64.linux.gnu' installed. Run:

      juliaup update

in your terminal shell to install Julia 1.12.7+0.x64.linux.gnu and update the 'release' channel to that version.

      | Documentation: https://docs.julialang.org
      | Type "?" for help, "]?" for Pkg help.
      | Version 1.12.5 (2026-02-09)
      | Official https://julialang.org release

julia> using Literate

julia> Literate.script("daisyworld-luminosity.jl", "docs")
[ Info: generating plain script file from '~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity.jl'
[ Info: writing result to '~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity.jl'
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity.jl"

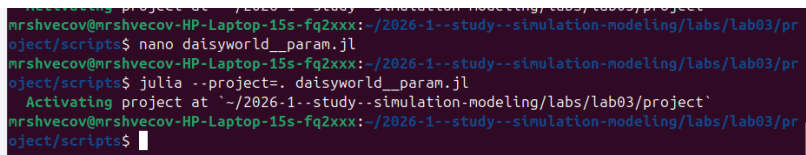
julia> Literate.markdown("daisyworld-luminosity.jl", "docs")
[ Info: generating markdown page from '~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity.jl'
[ Info: writing result to '~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity.md'
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity.md"
```

Выполнение скрипта scripts/daisyworld-luminosity.jl (рис. 13).

```
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr  
oject/scripts$ julia --project=. daisyworld-luminosity.jl  
  Activating project at `~/2026-1--study--simulation-modeling/labs/lab03/project`  
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr  
oject/scripts$
```

Рисунок 13: Выполнение скрипта:

Создание и выполнение scripts/daisyworld\_\_param.jl (рис. 14).



```
activating project at ~/2026-1--study--simulation-modeling/labs/lab03/project
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
ject/scripts$ nano daisyworld__param.jl
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
ject/scripts$ julia --project=. daisyworld__param.jl
  Activating project at `~/2026-1--study--simulation-modeling/labs/lab03/project`
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
ject/scripts$
```

Рисунок 14: Создание и выполнение scripts/daisyworld\_\_param.jl:

Создание литературного стиля `scripts/daisyworld__param.jl` (рис. 15).

```
object/scripts$ julia
Info The latest version of Julia in the 'release' channel is 1.12.7+0.x64.linux.gnu
. You currently have `1.12.5+0.x64.linux.gnu` installed. Run:

juliaup update

in your terminal shell to install Julia 1.12.7+0.x64.linux.gnu and update the `release` cha
nnel to that version.
```



```
| Documentation: https://docs.julialang.org
|
| Type "?" for help, "]"? for Pkg help.
|
| Version 1.12.5 (2026-02-09)
| Official https://julialang.org release
```

```
julia> using Literate

julia> Literate.script("daisyworld__param.jl", "docs")
[ Info: generating plain script file from `~/2026-1--study--simulation-modeling/labs/lab03/pro
ject/scripts/daisyworld__param.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/p
roject/scripts/docs/daisyworld__param.jl`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisywo
rld__param.jl"
```

```
julia> Literate.markdown("daisyworld__param.jl", "docs")
[ Info: generating markdown page from `~/2026-1--study--simulation-modeling/labs/lab03/proj
ect/scripts/daisyworld__param.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/p
roject/scripts/docs/daisyworld__param.md`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisywo
rld__param.md"
```

```
julia> Literate.notebook("daisyworld__param.jl", "docs")
[ Info: generating notebook from `~/2026-1--study--simulation-modeling/labs/lab03/project/s
cripts/daisyworld__param.jl`
```

Создание и выполнение scripts/daisyworld-count\_\_param.jl (рис. 16).

```
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$ nano daisyworld-count__param.jl
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$ julia --project=. daisyworld-count__param.jl
  Activating project at `~/2026-1--study--simulation-modeling/labs/lab03/project`
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$
```

Рисунок 16: Создание и выполнение scripts/daisyworld-count\_\_param.jl:

## Создание литературного стиля scripts/daisyworld-count\_\_param.jl (рис. 17).

```
julia> using Literate

julia> Literate.script("daisyworld-count__param.jl", "docs")
[ Info: generating plain script file from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count__param.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count__param.jl`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count__param.jl"

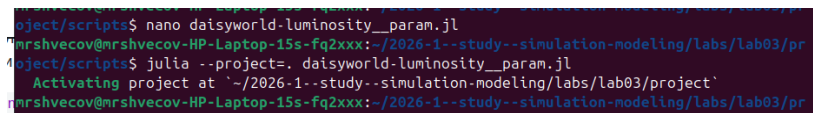
julia> Literate.markdown("daisyworld-count__param.jl", "docs")
[ Info: generating markdown page from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count__param.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count__param.md`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count__param.md"

julia> Literate.notebook("daisyworld-count__param.jl", "docs")
[ Info: generating notebook from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count__param.jl`
[ Info: executing notebook `daisyworld-count__param.ipynb`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count__param.ipynb`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count__param.ipynb"

julia>
```

Рисунок 17: Создание литературного стиля scripts/daisyworld-count\_\_param.jl:    

Создание и выполнение scripts/daisyworld-luminosity\_\_param.jl (рис. 18).



```
project/scripts$ nano daisyworld-luminosity__param.jl
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
project/scripts$ julia --project=. daisyworld-luminosity__param.jl
Activating project at `~/2026-1--study--simulation-modeling/labs/lab03/project`
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
```

Рисунок 18: Выполнение и создание :

## Создание литературного стиля scripts/daisyworld-luminosity\_\_param.jl (рис. 19).

```
julia> using Literate

julia> Literate.notebook("daisyworld-luminosity__param.jl", "docs")
[ Info: generating notebook from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity__param.jl`
[ Info: executing notebook `daisyworld-luminosity__param.ipynb`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity__param.ipynb`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity__param.ipynb"

julia> Literate.script("daisyworld-luminosity__param.jl", "docs")
[ Info: generating plain script file from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity__param.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity__param.jl`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity__param.jl"

julia> Literate.markdown("daisyworld-luminosity__param.jl", "docs")
[ Info: generating markdown page from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity__param.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity__param.md`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity__param.md"

julia>
```

Рисунок 19: Создание литературного стиля scripts/daisyworld-luminosity\_\_param.jl:



## Раздел 3

### Выводы

В данной лабораторной работе мы попрактиковались в создании литературного стиля кода на примере Модели Daisyworld.

